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BODY MOUNTING SYSTEM FOR A MODEL VEHICLE

Abstract

A body mounting assembly, model vehicle, and method for engaging a model vehicle body to a model vehicle chassis are provided. The body mounting assembly includes a latch assembly. The latch assembly further includes a latch member with a latch engagement surface. The body mounting assembly also includes a retainer assembly. The retainer assembly further includes a retainer engagement surface. The latch assembly is releasably engaged to the retainer assembly when the latch engagement surface and the retainer engagement surface are interlock together. A body mount able to be coupled to a model vehicle body includes one of the latch assembly or the retainer assembly and a chassis mount able to be coupled to the model vehicle chassis includes the other corresponding retainer assembly or the latch assembly corresponding to the body mount.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims the benefit of a related U.S. Provisional Application Ser. No. 63/127,662 filed Dec. 18, 2020, entitled “BODY MOUNT SYSTEM FOR A MODEL VEHICLE,” to Jonathan Scott WOOD, et al., and U.S. Pat. No. 11,911,708 B2, filed Dec. 16, 2021 to Jonathan Scott WOOD et al., and is a continuation of currently co-pending U.S. patent application Ser. No. 18/412,830, filed Jan. 15, 2024, entitled “BODY MOUNT SYSTEM FOR A MODEL VEHICLE,” to Jonathan Scott WOOD, et al., the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] The following descriptions and examples are not admitted to be prior art by virtue of their inclusion in this section.

[0003] Radio-Controlled or RC model vehicles are a popular hobby for a growing segment of the population. As the hobby has grown and expanded, the electronics have become more sophisticated and propulsion systems have expanded to include both electronically powered along with nitro or combustible fuel powered RC vehicles. Various factors such as the ease of operation, the run time of RC model vehicles, and the features and abilities of the RC model vehicles have increased dramatically to provide a more realistic operational environment. However, one area that may not have seen similar levels of development with regards to the ease of operation is the removal and attachment of a model vehicle body to a model vehicle chassis.

[0004] Traditional methods of removal and attachment of a model vehicle body have involved aligning a series of holes in a model vehicle body with a corresponding series of mounting posts attached to a model vehicle chassis. The mounting posts are extended through the holes from the interior of the model vehicle body. Small metal clips are then inserted through holes in the visible portions of the mounting post to secure the model vehicle body to the model vehicle chassis. This process takes time. In addition, a user has to be on guard against losing or misplacing any of the clips, oftentimes in an outdoor environment. Since the model vehicle body must be removed every time in order to activate the RC model vehicle or whenever there is a need to change or charge the batteries (such as with electrical propulsion systems), valuable run time is wasted performing this required and complex procedure.

SUMMARY

[0005] This summary is provided to introduce a selection of concepts that are further described below in the detailed description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0006] In accordance with one embodiment, a body mounting assembly is provided including a latch assembly comprising a latch member. The latch member includes a latch engagement surface. The body mounting assembly includes a retainer assembly comprising a retainer engagement surface. The latch assembly is releasably coupled to the retainer assembly when the latch engagement surface and the retainer engagement surface interlock together. Wherein a body mount

configured to be coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly and a chassis mount configured to be coupled to the model vehicle chassis comprises a corresponding one of the retainer assembly or the latch assembly corresponding to the body mount.

[0007] In another embodiment a model vehicle is provided that includes a body mounting assembly. The body mounting assembly includes a reinforcement member and a latch assembly. The latch assembly includes a latch member comprising a latch engagement surface. The body mounting assembly further includes a reinforcement retainer and a retainer assembly. The retainer assembly includes a retainer aperture comprising a retainer engagement surface.

[0008] The first side of the retainer aperture comprises the retainer engagement surface. The latch assembly is releasably coupled to the retainer assembly when the latch engagement surface and the retainer engagement surface interlock together. Further, the reinforcement member is constrained in at least one direction by the reinforcement retainer after assembly.

[0009] In addition, the body mounting assembly includes a body mount coupled to a model vehicle body that includes one of the latch assembly or the retainer assembly, and one of the reinforcement member or the reinforcement retainer. The body mounting assembly also includes a chassis mount coupled to a model vehicle chassis that includes a corresponding other of the retainer assembly or the latch assembly, and a corresponding other of the reinforcement retainer or the reinforcement member corresponding to the body mount.

[0010] In still another embodiment a method for securing a model vehicle body to a model vehicle chassis is provided. The method includes providing a latch assembly including a latch member comprising a latch engagement surface. The method also includes providing a retainer assembly that includes a retainer aperture. The retainer aperture includes a retainer engagement surface.

[0011] Still further, the method includes engaging the latch assembly and the retainer assembly, interlocking the latch engagement surface and the retainer engagement surface. Wherein a body mount coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly and wherein a chassis mount coupled to a model vehicle chassis comprises a corresponding other of the retainer assembly or the latch assembly corresponding to the body mount.

[0012] Other or alternative features will become apparent from the following description, from the drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Certain embodiments will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying drawings illustrate only the various implementations described herein and are not meant to limit the scope of various technologies described herein. The drawings are as follows:

[0014] FIG. 1 is a schematic assembly view of a model vehicle body and a model vehicle chassis incorporating a body mounting assembly, according to an embodiment of the current disclosure;

[0015] FIG. 2 is an enlarged partial assembly view of a latch assembly and a retainer assembly prior to assembly of a body mounting assembly, according to an embodiment of the current disclosure;

[0016] FIG. 3 is an enlarged sectional view of the latch assembly and the retainer assembly of FIG. 2 after assembly of a body mounting assembly, according to an embodiment of the current disclosure;

[0017] FIG. 4 is a schematic assembly view of a model vehicle body and a model vehicle chassis

incorporating a body mounting system, according to another embodiment of the current disclosure; [0018] FIG. 5 is another schematic assembly view of the model vehicle body and the model vehicle chassis of FIG. 4, but showing the model vehicle body from a lower view, according to another embodiment of the current disclosure;

[0019] FIG. 6 is a schematic assembly view of the second body mounting assembly comprising a body mount and a chassis mount of the body mounting assembly of FIGS. 4 and 5, shown prior to assembly, according to another embodiment of the current disclosure;

[0020] FIG. 7 is an enlarged schematic perspective view of the body mount of FIG. 6 shown from a lower perspective, according to another embodiment of the current disclosure;

[0021] FIG. 8 is an enlarged partial view of the chassis of FIGS. 4 and 5 showing the chassis mount of FIG. 6 attached to the model vehicle chassis, according to another embodiment of the current disclosure; and

[0022] FIG. 9 is an enlarged sectional view of the body mount and the chassis mount of FIG. 6 shown in an assembled position, according to another embodiment of the current disclosure.

DETAILED DESCRIPTION

[0023] In the following specification, numerous specific details are set forth to provide a thorough understanding of embodiments of the present disclosure. However, those skilled in the art will appreciate that the embodiments may be practiced without such specific details. In other instances, well-known elements have been illustrated in schematic or block diagram form in order not to obscure embodiments of the present disclosure in unnecessary detail.

[0024] Reference throughout the specification to “one embodiment,” “an embodiment,” “some embodiments,” “one aspect,” “an aspect,” or “some aspects” means that a particular feature, structure, method, or characteristic described in connection with the embodiment or aspect is included in at least one embodiment of the present disclosure. Thus, the appearance of the phrases “in one embodiment” or “in an embodiment” or “in some embodiments” in various places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, methods, or characteristics may be combined in any suitable manner in one or more embodiments. The words “including,” “comprising,” “containing” and “having” shall have the same meaning as the word “comprising.”

[0025] Moreover, inventive aspects lie in less than all features of a single disclosed embodiment. Thus, the claims following the Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment.

[0026] Radio Controlled (RC) model vehicles usually comprise a scale version of a model vehicle body coupled to a model vehicle chassis. The model vehicle chassis contains the electronics and servos required for operating an RC model vehicle. The propulsion systems could be a combustion engine (e.g., nitro powered engine) or electric motor (e.g., dc battery powered motor). In most cases, the model vehicle body must be removed in order to activate and deactivate the Electronic Speed Control (ESC), or in the case of electric propulsion, to charge, replace, or connect a battery pack. Conversely, during operation, the model vehicle body must remain securely coupled to the model vehicle chassis while the vehicle is run under a variety of conditions and circumstances.

[0027] Previous versions of RC model vehicles would typically include mounting posts, coupled to a model vehicle chassis, with protruding features located at the top of each of the posts extending through the model vehicle body. Each of the protruding features may include holes to attach individual clips, for example. The model vehicle body is lowered onto the model vehicle chassis, allowing the protruding features to extend above an exterior surface of the model vehicle body. The model vehicle body rests upon body support pads coupled to the mounting posts. Then a number of small clips are inserted through the holes in the protruding features to secure the model vehicle in place between the clips and the body support pads.

[0028] When the model vehicle body is removed from the model vehicle chassis, sometimes at a track or in the field, a few of the many clips may be lost or misplaced. This results in frustration

and an inability to fully secure the model vehicle body afterwards unless a supply of spare clips is maintained. In addition, inserting and removing each of the clips into the holes at the top of the mounting posts requires precision and time. The precision is primarily needed for inserting clips into the holes, but time is required for both the insertion and removal of clips from each of the mounting posts. Accordingly, the entire process for removing and replacing the model vehicle body takes a relatively long time to perform and comes with the risk of losing one or more of the many clips needed to fully secure the model vehicle body.

[0029] Referring generally to FIG. 1, this figure shows an embodiment of a model vehicle **100** including a transparent model vehicle body **200** and a model vehicle chassis **300**. In this illustrative embodiment, the model vehicle **200** is coupled to a first body mounting assembly **800**, comprising a first body mount **600** and a corresponding first chassis mount **400**. In addition, the model vehicle body **200** is coupled to a second body mounting assembly **900**, comprising a second body mount **700** and a corresponding second chassis mount **500**.

[0030] In this particular embodiment, two body mounting assemblies **800,900** are shown mounted transversely in the model vehicle body **200** and the model vehicle chassis **300**. They could equivalently be mounted longitudinally in the model vehicle body **200** and the model vehicle chassis **300**. While two body mounting assemblies **800,900** are shown in this illustrative example, there could be one or three or more in other embodiments according to application.

[0031] The first body mounting assembly **800** is shown near the model vehicle's **100** body front **240** and chassis front **340**. The second body mounting assembly **900** is shown near the model vehicle's **100** body rear **250** and chassis rear **350**. However, in other embodiments the first and second body mounting assemblies **800, 900** could alternatively be in the middle or along the model vehicle's **100** body left side **260** and chassis left side **360** and/or the model vehicle's **100** body right side **270** and chassis right side **370**. Appropriate placement and numbers of body mounting assemblies **800,900** may vary according to model vehicle type and/or configuration.

[0032] Looking at FIG. 2, this generally representative example shows an enlarged schematic portion of an assembly view of the first body mounting assembly **800**'s first body mount **600** and first chassis mount **400** in an unassembled condition. The first body mounting assembly **800** will be described in detail as an exemplary embodiment of a body mounting assembly. Only the first body mounting assembly **800** will be discussed in the interest of reducing redundancy. And although 'first' nomenclature will be used to signify the first body mounting assembly **800** components, the detailed description should be read as though the specific 'first' was not present and that the description applied to a general description of an embodiment of the component following that prefix.

[0033] Turning to FIG. 3, this exemplary illustration shows an enlarged cross-sectional view of a portion of the first body mounting assembly **800** of FIG. 2 in an assembled condition. In this case, first body mount **600** and first chassis mount **400** are shown interlocked together. In this illustration, a model vehicle body **200** has been secured to a model vehicle chassis **300** even though the portions of the model vehicle body **200** and the model vehicle chassis **300** are not shown in order to aid in clarification of the description (refer to FIG. 1 for the model vehicle body **200** and the model vehicle chassis **300** in an unassembled state).

[0034] The chassis mount **400** includes a first retainer assembly **410** configured for accepting and locking with a corresponding structure of the first body mount **600**. The first retainer assembly **410** includes a first retainer aperture **420** and a first retainer engagement surface **430**. The first retainer engagement surface **430** is provided on a first side of the first retainer aperture **440**. In addition to the first retainer engagement surface **430**, the first side of the first retainer aperture **440** also includes a first retainer lead-in ramp **435**. The first retainer lead-in ramp **435** is angled downward to the right into the first retainer aperture **420**. The first retainer lead-in ramp **435** is configured to facilitate alignment and assembly of the first body mount **600** and the first chassis mount **400**.

[0035] The first retainer engagement surface **430** also extends into the first retainer aperture **410**

and is angled downward and to the right, as seen in the figure. While the first retainer lead-in ramp **435** is configured to facilitate alignment and assembly of the first body mount **600** and the first chassis mount **400**, the first retainer engagement surface **430** is configured to securely interlock with a corresponding feature of the first body mount **600**. Securely interlocking the first body mount **600** to the first chassis mount **400** inhibits or prevents inadvertent or unintentional separation of the model vehicle body **200** from the model vehicle chassis **300** and will be explained later in more detail.

[0036] The first retainer aperture **420** further includes a second side of the first retainer aperture **450**, opposite to the first side of the first retainer aperture **440**. The first retainer aperture **420** is shown in FIGS. **1** and **2** as substantially square so there are two additional sides to the first retainer aperture **420**. The substantially square first retainer aperture **420** is configured to accept a correspondingly configured first latch assembly **610**. While a square is used in this illustrative example, other embodiments may use other configurations or geometric shapes as appropriate for the application. For example, a triangular, circular, or other geometric shape may also be used instead of a square for the first retainer aperture **420**. Whatever shape is used in an embodiment should accommodate and correspond to the overall outer boundary of the associated component in first body mount **600**.

[0037] The first body mount **600** is coupled to the model vehicle body **200** and is illustrated as comprising a first latch assembly **610**. However, in other embodiments, the first body mount **600** may comprise a first retainer assembly **410**. In still other embodiments, the first body mount **600** may comprise combinations of a first latch assembly **610** and a first retainer assembly **410**. The first latch assembly **610** includes a first latch member **620** comprising a first latch engagement surface **630**. In addition, the first latch member **620** comprises a first latch release **640** and a first latch support **650**. In embodiments in which a second latch assembly is included in the body mount **600** (see FIG. **2**), a latch assembly connecting member **660** may be used to connect the first latch assembly **610** to the second latch assembly. In some embodiments, the latch assembly connecting member **660** may transversely or longitudinally span across the model vehicle **200**.

[0038] The first latch member **620** is illustrated in this exemplary figure as a resilient U-shaped cantilever snap fit latch. However, other latch designs can be used such as cantilever snap fit latches and L shaped cantilever snap fit latches, for example. Further, the U-shaped cantilever snap fit first latch member **620** uses a first latch engagement surface **630** that is an angled protrusion configured to interlock with a retainer engagement surface **430** that is also an angled protrusion in the first chassis mount **400**. Both of the engagement surfaces (**430**, **630**) are shown at a negative angle (i.e., a negative return angle) to increase the retention ability of the interlocking features.

[0039] In most cases of snap fit latch design, the first engagement surface **630** is either perpendicular or at a positive angle (i.e., a positive return angle) to the rest of the first latch member **620**. With a perpendicular angle or a positive angle for the first engagement surface **630**, the application of an increasing vertical separation force between the model vehicle body **200** and the model vehicle chassis **300** may continue until a point at which the latch member slides over the corresponding interlocking feature. Generally, the latch assembly and corresponding retaining assembly are disengaged without failure of the components in each.

[0040] With a negative engagement angle as shown for the first latch member **620**, an increasing vertical separation force between the model vehicle body **200** and the model vehicle chassis **300** may result in the failure of one or both of the components in a first latch assembly **610** and/or a first retainer assembly **410**. While typical snap fit interlocking latches may be disengaged through the application of a separation force, snap fit interlocking latches with negative engagement angles must be physically disengaged prior to application of a separation force. Application of a separation force between the model vehicle body **200** and the model vehicle chassis **300** would draw the first latch engagement surface **630** and the first retainer engagement surface **430** into a tighter interlocking position usually until a failure occurred.

[0041] In order to disengage the first latch assembly **610** from the first retainer assembly, the first latch assembly **610** further includes a first latch release **640**. Moving the first latch release member **640** prior to the application of a vertical separation force provides a way to disengage the model vehicle body **200** from the model vehicle chassis **300** without damage. The slanted surface of the first latch release **640** may be at a positive angle (i.e., a positive lead angle) in order to facilitate assembly. As the first latch assembly **610** is inserted into the first retainer assembly **410**, first latch release member **640** may slide against the first retainer lead-in ramp **435**, also shown at a positive angle (i.e., a positive lead angle). Accordingly, the first latch member **620** is moved to the right of the illustration and then into position within the first retainer assembly **410**.

[0042] A first latch support **650** abuts the second side of the first retainer aperture **450**, guiding the rest of the first latch assembly **610** into position for assembly. As the first latch support **650** is inhibited from moving further to the right and as the first latch release member **640** slides against the first retainer lead-in ramp **435**, the first latch member **620** resiliently bends to the right as well. When the trailing edge of the first latch release member **640** passes the trailing edge of the first retainer lead-in ramp **435**, the first latch member **620** resiliently moves to the left, interlocking together the first latch engagement surface **630** and the first retainer engagement surface **430**.

[0043] In order to disengage the first latch engagement surface **630** and the first retainer engagement surface **430**, the first latch release member **640** is operated or moved closer to the first latch support **650**, away from the first retainer engagement surface **430**. This allows the first latch engagement surface **630** to vertically clear the first retainer engagement surface **430**. While clear, the model vehicle body **200** can be moved vertically to separate from the model vehicle chassis **300**. In some cases, the first latch release member **640** is initially operated to clear the first latch engagement surface **630** and operated or moved vertically to disengage the first latch assembly **610** from the first retainer assembly **410**, allowing the first body mount **600** to disengage from the first chassis mount **400**.

[0044] Operating the first latch release member **640** involves opposing the resilient bias produced by the first latch member **620**. The resilient bias is in the direction of the first retainer engagement surface **430** during assembly. The bias helps to position the first latch engagement surface **630** in a position to lock with the first retainer engagement surface **430**. In addition, the bias may function to keep the first latch engagement member **630** interlocked and engaged with the first retainer engagement member **430** during operation.

[0045] In some embodiments in the assembled position shown in FIG. 3, the resilient bias may be zero. However, during assembly when the first release surface **640** slides to the right due to the first retainer lead-in ramp **435** as the first latch member **620** is inserted into the first retainer aperture **420**, the resilient bias is built up in the direction of the first retainer engagement surface **430**. While in other embodiments, the resilient bias will keep the portion of the first latch member **620** firmly against the left side of the first retainer aperture **420** during normal operation.

[0046] As shown in exemplary FIG. 1, this particular embodiment of the current disclosure shows a model vehicle **100** comprising a first body mounting assembly **800** and a second body mounting assembly **900**. Again, while two transversely mounted body mounting assemblies **800**, **900** are shown, other embodiments may comprise only a single body mounting assembly or three or more body mounting assemblies. The one or more body mounting assemblies **800**, **900** may further be mounted longitudinally or in any orientation applicable for the intended application in still other embodiments.

[0047] In addition, while the first latch assembly **610** is shown in the first body mount **600** and the first retainer assembly **410** is shown in the first chassis mount **400**, the numbers of assemblies (both body and retainer) as well as the location of assemblies (body mount and chassis mount) can vary according to embodiment and application. In some cases, the first latch assembly **610** could be in the first chassis mount **400** and the first retainer assembly **410** could be in the first body mount **600**. While in still other embodiments, the first body mount **600** could also comprise both a first latch

assembly **610** and a first retainer assembly **410**, and the first chassis mount **400** would comprise a corresponding alternative set of a first retainer assembly **410** and first latch assembly **610**. And in further embodiments, the first body mount **600** may comprise one or more first latch assemblies **610** and one or more first retainer assemblies **410** while the first chassis mount **400** comprises an alternative set of one or more first retainer assemblies **410** and one or more first latch assemblies **610**.

[0048] As shown in the illustrative embodiment in FIG. 1, the first body mount **600** of the first body mounting assembly **800** and the second body mount **700** of the second body mounting assembly **700** are rigidly coupled to an internal body frame **230**. The internal body frame **230** is further coupled to an interior surface of the model vehicle body **200**. However, in other embodiments the first body mount **600** and the second body mount **700** may be rigidly coupled directly to the interior surface of the model vehicle body **200**. While in still other embodiments, there may be a single internal body frame **230** extending the length of the model vehicle **100** or in other cases, two or more internal body frames such as one for the front of the model vehicle **240** and one for the rear of the model vehicle **250**, for example.

[0049] FIG. 1 shows the first chassis mount **400** and the second chassis mount **500** are shown having implemented the first and second retainer assemblies **410**, **510** at all for wheel wells **310A-D**. This creates 2 latching and retaining assemblies per side of the model vehicle **100**. The model vehicle front—**340**, the model vehicle rear—**350**, the model vehicle left—**360** and the model vehicle right **370** may all have redundant body mounting assemblies. Of course, this embodiment and implementation are for illustrative purposes only and other locations and numbers of latching and retaining assemblies may differ according to application.

[0050] Referring generally to FIGS. 4 and 5, these illustrative schematic assembly drawings show a model vehicle **1000** comprising a model vehicle body **1200** and a model vehicle chassis **1300** incorporating an embodiment of the current disclosure. In this exemplary embodiment, a first body mounting assembly **1800** and a second body mounting assembly **1900** are shown. The first body mounting assembly **1800** comprises the first body mount **1600** and the first chassis mount **1400**. The second body mounting assembly **1900** comprises the second body mount **1700** and the second chassis mount **1500**. In FIG. 5, the model vehicle body **1200** illustrates from a lower perspective view which shows a different perspective of the first body mount **1600** and the second body mount **1700**.

[0051] Turning now to FIG. 6, this illustrative figure shows an enlarged perspective schematic view of the second body mounting assembly **1900** of FIGS. 4 and 5 without the model vehicle body **1200** or the model vehicle chassis **1300** of the model vehicle **1000** in the interest of increasing clarity. The second body mount **1700** and the second chassis mount **1500** are shown in an unassembled state. In order to assemble the two mounts **1500**, **1700**, the second body mount **1700** would be brought lower until it engages the second chassis mount **1500**. The assembly would continue until a second latching assembly is able to engage a second retainer assembly (discussed later). With the second latching assembly engaged with the second retainer assembly, the model vehicle body **1200** is secured to the model vehicle chassis **1300** and the model vehicle **1000** is ready for operation.

[0052] Referring to FIG. 7, this exemplary illustration shows an enlarged lower perspective schematic view of the second body mount **1700** of FIG. 6. As shown in this view, the second body mount **1700** comprises a second latch assembly **1710**. However, in other embodiments the second body mount **1700** may comprise two or more second latch assemblies **1710**. Comparing the first body mount **1600** and the second body mount **1700** in FIG. 5, both the first and the second body mounts **1600**, **1700** have a single latch assembly (**1610**, **1710**). However, the number of latch assemblies **1610**, **1710** does not have to be equal to one another in each of the body mounts **1600**, **1700**. In addition, one body mount (**1600**, **1700**) may include a latch assembly (**1610**, **1710**) while the other body mount (**1700**, **1600**) may include a retainer assembly (**1510**, **1410**).

[0053] In some embodiments, the body mounts (**1600**, **1700**) may comprise different numbers and combinations of latch assemblies (**1610**, **1710**) and/or retainer assemblies (**1510**, **1410**) as appropriate for a particular embodiment. In some cases, the second body mount **1700** may comprise both a second latch assembly **1710** and a second retainer assembly **1510** while the second chassis mount **1500** comprises a corresponding set of a second retainer assembly **1510** and a second latch assembly **1710**.

[0054] As shown, this embodiment of the second latch assembly **1710** includes a second latch support **1714**, a second latch member **1720** with a second latch resilient arch **1722**, a second latch engagement surface **1730**, and a second latch release **1740**. As with the first embodiment, the second latch engagement surface **1730** is shown at a negative angle (i.e., a negative return angle) while the second latch release **1740** is shown at a positive angle (i.e., a positive lead angle). This illustrative example of a second latch assembly **1710** comprises a resilient U-shaped cantilever snap-fit latch. However, other configurations of snap fit latches can be used such as a cantilever snap-fit latch and an L-shaped cantilever snap-fit latch, among others according to the requirements of the application.

[0055] The use of a second latch resilient arch **1722** allows for very large deflections of the end portion (i.e., distal portion) of the second latch member **1720** that comprises the second engagement surface **1730** and the second latch release **1740** (also refer to FIG. 9). The second latch resilient arch **1722** facilitates the very large deflections without inducing high strains at the base of the second latch member **1720**. Both U-shaped cantilever snap-fit latches and L-shaped cantilever snap-fit latches allow for a reduction in strain caused by the deflection as compared to a typical cantilever snap-fit latch.

[0056] The second latch assembly **1710** is shown in this illustrative embodiment with a substantially square overall horizontal circumference. This configuration matches the substantially square overall horizontal circumference of the second retainer assembly **1510** (explained in more detail later). By having a substantially square overall horizontal circumference, primary forces between the model vehicle body **1200** and the model vehicle chassis **1300** along an x-axis and y-axis are restrained by corresponding flat sides of the substantially square configuration, distributing the force along the sides. However, in other embodiments other geometric configurations may be used depending upon the application. For example, substantially circular and triangular horizontal circumferences among others may provide a more appropriate configuration for an embodiment.

[0057] In addition to the second latch assembly **1710**, this embodiment of the second body mount **1700** further comprises second reinforcement members **1712A**, **1712B**. The second reinforcement members **1712A**, **1712B** provide additional support along at least one geometric axis to an assembled second body mounting assembly **1900**. In this exemplary embodiment, the second reinforcement members **1712A**, **1712B** provides additional support along two perpendicular horizontal directions, i.e., the x and y axis.

[0058] Including the second reinforcement members **1712A**, **1712B** in the second body mounting assembly **1900** may reduce or inhibit some of the resultant forces generated during operation of the model vehicle **1000** that act on the second retainer assembly **1510** and the second latch assembly **1710**, between the model vehicle body **1200** and the model vehicle chassis **1300**. In addition, the length of the second reinforcement members **1712A**, **1712B** may also function as a guide to facilitate placing the model vehicle body **1200** in the proper position relative to the model vehicle chassis **1300** during assembly. In some cases, the distal end of the second reinforcement members **1712A**, **1712B** may be tapered to aid insertion into the second reinforcement retainers **1512A**, **1512B**. The length of the second reinforcement members **1712A**, **1712B** may also help to keep the model vehicle body **1200** in place during a failure of either the second retainer assembly **1510** and/or the second latch assembly **1710**.

[0059] In this embodiment, the second reinforcement members **1712A**, **1712B** are shown as extended rectangular protrusions with a substantially square perimeter as a horizontal cross-section

(i.e., actual cross-section is in the form of an I-beam). However, other horizontal configurations may be used as second reinforcement members **1712A**, **1712B**, according to the needs of an embodiment or application. For example, the protrusions may be cylindrical with circular horizontal cross-sections or even comprising triangular or other combinations of polygonal, or arcuate shaped horizontal cross-sections. As with the second latch support **1714**, the distal (i.e., the lower as seen in the figure) end of the second reinforcement members **1712A**, **1712B** may be narrowed with angled surfaces to facilitate assembly and mating with corresponding second reinforcement retainers **1512A**, **1512B** (see FIG. 8) of the second chassis mount **1500**.

[0060] Referring now to FIG. 8, this figure is an exemplary illustration of an enlarged partial schematic assembly view of the second chassis mount **1500** as shown in FIGS. 4 and 5. In this view, the second retainer assembly **1510** is visible showing the top of the second retainer aperture **1520** and the second reinforcement retainers **1512A**, **1512B**. As with other components, the leading entry into the upper portions (as seen in this view) of the second retainer aperture **1520** and the second reinforcement retainers **1512A**, **1512B**, are enlarged with angled surfaces. The angled surfaces facilitate assembly, guidance, and engagement with the second latch assembly **1710** and the second reinforcement members **1712A**, **1712B** of the second body mount **1700**.

[0061] Turning generally to FIG. 9, the embodiment of the second body mounting assembly **1900** is shown in an assembled configuration in which the second body mount **1700** and the second chassis mount **1500** are engaged or interlocked together. In this view, the remaining structure and individual components of the second retainer assembly **1510** can be seen. As shown, the second retainer assembly **1510** comprises a second retainer aperture **1520** including a first side of the second retainer aperture **1516** and a second side of the second retainer aperture **1514**.

[0062] The first side of the second retainer aperture **1516** comprises a second retainer lead-in ramp **1535** that moves the lower (or distal) portion of the second latch member **1720** to the right (as seen in the figure), as the second latch release member **1740** slides over the second retainer lead-in ramp **1535** during assembly. This results in the generation of a resilient force urging the lower portion of the second latch member **1720** to the left (as seen in the figure). When the highest portion of the second latch release member **1740** clears the lowest portion of the second retainer lead-in ramp **1535**, the lower portion of the second latch member **1720** resiliently moves to the left, resulting in the second latch engagement surface **1730** opposing the second retainer engagement surface **1530**.

[0063] The second latch engagement surface **1730** and the second retainer engagement surface **1530** each have a negative engagement angle or a negative return angle. Since both engagement surfaces **1530**, **1730** have a negative engagement angle, attempting to vertically remove the model vehicle body **1200** from the model vehicle chassis **1300** will result in the lower portion of the second latch member **1720** being drawn to the leftmost position in the second retainer aperture **1520**, furthering the engagement between the second latch engagement surface **1730** and the second retainer engagement surface **1530**. Unlike other snap-fit latch designs, application of additional separation force will eventually result in failure of the second latch assembly **1710**, the second retainer assembly **1510**, or both.

[0064] Application of a horizontal force upon the second latch release member **1740** to the right, moves the second latch engagement surface **1730** out of engagement or interlock with the second retainer engagement surface **1530**. Application of a vertical force to separate the model vehicle body **1200** from the model vehicle chassis **1300** then results in disengagement of the second body mount **1700** from the second chassis mount **1500** and upon disengagement of all body mounting assemblies (**1800**, **1900**, etc.), the model vehicle body **1200** is removable from the model vehicle chassis **1300**.

[0065] During assembly, the second retainer lead-in ramp **1535** moves the lower portion of the second latch member **1720** to the right. As shown in FIG. 9, the base of the second latch member **1720** is inhibited or resisted from moving right by a corresponding amount due to the interaction of the second latch support member **1714** and the second side of the second retainer aperture **1540**.

The second latch support member **1714** abuts the second side of the second retainer aperture **1514** preventing further horizontal translation of the second latch assembly **1710** relative to the second retainer assembly **1510**. In addition, the interaction of the second reinforcement members **1712A**, **1712B** and the second reinforcement retainers **1512A**, **1512B** also inhibit or restrict horizontal movement of the second latch assembly **1710** relative to the second retainer assembly **1510**. [0066] In some embodiments, the interlocking second latch engagement surface **1730** and the second retainer engagement surface **1530** may function to resist the vertical separation of the model vehicle body **1200** from the model vehicle chassis **1300**. The second latch support member **1714** abutting the sides of the second retainer aperture **1520** and/or the second reinforcement members **1712A**, **1712B** abutting the sides of the second reinforcement retainers **1512A**, **1512B** may function to inhibit or restrain the horizontal motion of the model vehicle body **1200** relative to the model vehicle chassis **1300**. While the second latch support member **1714** sliding against the sides of the second retainer aperture **1520** and/or the second reinforcement members **1712A**, **1712B** sliding against the second reinforcement retainers **1512A**, **1512B** may also function to facilitate guiding the model vehicle body **1200** to the appropriate mounting location or position relative to the model vehicle chassis **1300**.

[0067] The inhibition or restriction of the second latch assembly **1710** moving horizontally relative to the second retainer assembly **1510** while the lower portion of the second latch member **1720** moving to the right (as seen in FIG. 9) due to the second retainer lead-in ramp **1535** during assembly, results in at least some of the resilient force being generated to urge the lower portion of the second latch member **1720** to the left (as seen in the figure). Appropriate design of the second latch member **1720** can result in a generated resilient force strong enough to resist disengagement during operation of the model vehicle **1000** while still allowing for a reasonable disassembly or disengagement force.

[0068] Still another illustrative embodiment includes a method for securing a model vehicle body to a model vehicle chassis. This method involves providing a latch assembly that includes a latch member. The latch member further includes a latch engagement surface. Also, the method includes providing a retainer assembly that includes a retainer aperture. The retainer aperture includes a retainer engagement surface.

[0069] The method may further include engaging the latch assembly and the retainer assembly, interlocking the latch engagement surface and the retainer engagement surface. Wherein a body mount coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly. Also, wherein a chassis mount coupled to a model vehicle chassis comprises a corresponding other of the retainer assembly or the latch assembly corresponding to the body mount. Still further, wherein the latch engagement surface and the retainer engagement surface comprise negative engagement angles.

[0070] Some embodiments of the method may further include providing a latch release member on the latch member and moving the latch release member in a horizontal direction to decouple the latch assembly and the retainer assembly from engagement after assembly. In addition, embodiments may include providing a reinforcement member and a reinforcement retainer, wherein the body mount includes one of the reinforcement member or the reinforcement retainer. Also, wherein the chassis mount further includes a corresponding other of the reinforcement retainer or the reinforcement member, and wherein the reinforcement member is constrained in at least one direction by the reinforcement retainer after assembly.

[0071] Elements of the embodiments have been introduced with either the articles “a” or “an.” The articles are intended to mean that there are one or more of the elements. The terms “including” and “having” are intended to be inclusive such that there may be additional elements other than the elements listed. The term “or” when used with a list of at least two elements is intended to mean any element or combination of elements.

[0072] Although only a few example embodiments have been described in detail above, those

skilled in the art will readily appreciate that a wide range of variations, modifications, changes, and substitutions are contemplated in the foregoing disclosure and, in some instances, some features of the present disclosure may be employed without a corresponding use of the other features.

[0073] In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents, but also equivalent structures. Thus, although a nail and a screw may not be structural equivalents in that a nail employs a cylindrical surface to secure wooden parts together, whereas a screw employs a helical surface, in the environment of fastening wooden parts, a nail and a screw may be equivalent structures.

[0074] It is the express intention of the applicant not to invoke 35 U.S.C. § 112, paragraph 6 for any limitations of any of the claims herein, except for those in which the claim expressly uses the words 'means for' together with an associated function.

Claims

1. A body mounting assembly comprising: a latch assembly comprising: a latch member resiliently coupled to the latch assembly comprising: a latch engagement surface; a retainer assembly comprising: a retainer engagement surface; wherein the latch assembly is releasably coupled to the retainer assembly when the latch engagement surface and the retainer engagement surface interlock together; wherein the latch member is resiliently biased against release of the latch member assembly from the retainer assembly; wherein a body mount configured to be coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly; and wherein a chassis mount configured to be coupled to the model vehicle chassis comprises a corresponding one of the retainer assembly or the latch assembly corresponding to the body mount.
2. The body mounting assembly according to claim 1, wherein the latch member further comprises: a latch release member; and wherein the latch assembly is released from the retainer assembly via operation of the latch release member.
3. The body mounting assembly according to claim 2, wherein the retainer assembly further comprises: a retainer aperture configured to accommodate at least a portion of the latch member; and wherein the retainer engagement surface is provided on a first side of the retainer aperture.
4. The body mounting assembly according to claim 3, wherein operation of the latch release member comprises application of a force directed against the latch release member in a first direction.
5. The body mounting assembly according to claim 4, wherein the latch member is resiliently coupled to the latch assembly via a U-shaped cantilever biased in a second direction opposite to the first direction used to operate the latch release member.
6. The body mounting assembly according to claim 1, wherein the body mount is provided proximate to a first body wheel well and the chassis mount is provided proximate to a first chassis wheel well corresponding to the first body wheel well.
7. The body mounting assembly according to claim 1, further comprising: a second latch assembly comprising: a second latch member comprising: a second latch engagement surface; a second retainer assembly comprising: a second retainer engagement surface; a second latch member comprising: a second latch engagement surface; wherein the second latch assembly is releasably coupled to the second retainer assembly via the second latch engagement surface interlocking with the second retainer engagement surface; wherein the body mount further comprises one of the second latch assembly or the second retainer assembly; and wherein the chassis mount further comprises a corresponding one of the second retainer assembly or the second latch assembly corresponding to the body mount.
8. The body mounting assembly according to claim 7, wherein the one of the latch assembly or the retainer assembly of the body mount is rigidly coupled to the one of the second latch assembly or

the second retainer assembly of the body mount via a latch assembly connecting member.

9. The body mounting assembly according to claim 7, wherein the one of the latch assembly or the retainer assembly of the body mount is configured to be provided proximate to a first body wheel well in the model vehicle body and the corresponding one of the retainer assembly or the latch assembly of the chassis mount is configured to be provided proximate to a first chassis wheel well in the model vehicle chassis corresponding to the first body wheel well; and wherein the one of the second latch assembly or the second retainer assembly of the body mount is configured to be provided proximate to a second body wheel well in the model vehicle body and the corresponding one of the second retainer assembly or the second latch assembly is configured to be provided proximate to a second chassis wheel well provided in the model vehicle chassis corresponding to the first body wheel well.

10. The body mounting assembly according to claim 9, wherein the first and the second body wheel wells and the first and the second chassis wheel wells are provided on a first model vehicle side.

11. The body mounting assembly according to claim 10, wherein the first model vehicle side is a front of the model vehicle.

12. The body mounting assembly according to claim 4, wherein the latch assembly further comprises a latch support; and wherein the latch support abuts a second side of the retainer aperture opposite the first side of the retainer aperture.

13. The body mounting assembly according to claim 3, wherein one of the body mount or the chassis mount further comprises a reinforcement member; wherein another of the corresponding chassis mount or the body mount further comprises a reinforcement retainer; and wherein the reinforcement member is constrained in at least one direction by the reinforcement retainer after assembly.

14. A model vehicle comprising a body mounting assembly comprising: a reinforcement member; a reinforcement retainer; a latch assembly comprising: a latch member resiliently coupled to the latch assembly comprising: a latch engagement surface; a retainer assembly comprising: a retainer aperture comprising: a retainer engagement surface; wherein a first side of the retainer aperture comprises the retainer engagement surface; wherein the latch assembly is releasably coupled to the retainer assembly when the latch engagement surface and the retainer engagement surface interlock together; wherein the latch member is resiliently biased against release of the latch member assembly from the retainer assembly; wherein the reinforcement member is constrained in at least one direction by the reinforcement retainer after assembly; wherein a body mount coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly and one of the reinforcement member or the reinforcement retainer; and wherein a chassis mount coupled to a model vehicle chassis comprises a corresponding other of the retainer assembly or the latch assembly and a corresponding other of the reinforcement retainer and the reinforcement member corresponding to the body mount.

15. The model vehicle according to claim 14, wherein the chassis mount and the body mount are provided proximate to a first model vehicle side.

16. The model vehicle according to claim 14, wherein the latch member further comprises: a latch release member; and wherein the latch assembly is released from the retainer assembly via operation of the latch release member.

17. The body mounting assembly according to claim 16, wherein the latch member is resiliently coupled to the latch assembly via a U-shaped cantilever biased to oppose a direction used to operate the latch release member.

18. A body mounting assembly comprising: a latch assembly comprising: a latch member resiliently coupled to the latch assembly comprising: a latch engagement surface; a latch release member; a retainer assembly comprising: a retainer engagement surface; wherein the latch assembly is releasably coupled to the retainer assembly when the latch engagement surface and the retainer engagement surface interlock together; wherein the latch member is resiliently biased

against release of the latch member assembly from the retainer assembly; wherein a body mount configured to be coupled to a model vehicle body comprises one of the latch assembly or the retainer assembly; wherein a chassis mount configured to be coupled to the model vehicle chassis comprises a corresponding one of the retainer assembly or the latch assembly corresponding to the body mount; and wherein the latch assembly is released from the retainer assembly via operation of the latch release member.

19. The body mounting assembly according to claim 18, wherein operation of the latch release member comprises application of a force directed against the latch release member in a first direction.

20. The body mounting assembly according to claim 19, wherein the latch member is resiliently coupled to the latch assembly via a U-shaped cantilever biased in a second direction opposite to the first direction used to operate the latch release member.
